

1 Split once

Stratified 80% development / 20% test split.

Lock the test sample before screening, tuning, or stacking.

2 Outer cross-fitting

Five development folds generate out-of-fold base-model probabilities.

Each participant is assessed by a model that did not use that participant.

3 Inner model design

Within each outer analysis set: three-fold tuning.

In every inner split, limma is fitted only on the inner analysis fold and applied to its assessment fold.

4 Outer refit + OOF

Choose hyperparameters; rerun limma on the full outer analysis set.

Fit the base model and predict the untouched outer assessment fold.

5 Learn stack

Concatenate cross-fitted base probabilities across outer folds.

Fit ridge-logistic stacking weights using these development-only OOF predictions.

6 Final base models on all development subjects

Run a fresh five-fold tuning exercise within the 80% development sample; limma is again learned separately in each tuning analysis fold.

After hyperparameters are chosen, rerun limma once on all 80% to obtain the final deployable feature set; refit each base model on all 80%.

7 One-time locked-test evaluation

Generate final base-model probabilities for the 20% test sample.

Apply the fixed ridge stacking coefficients learned from development OOF predictions.

Compute ROC/AUC with death as the fixed positive event; do not retune or rescreen.